NOSQL ASSIGNMENT

1. Create a new database named CDAC and connect to it.

```
mongosh mongodb://127.0.0.
[d5_96525_Deven@sales]>CREATE INDEX cnum_odate_idx
    -> ON orders(cnum,odate);
Query OK, 0 rows affected (0.733 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>exit;
Bye

C:\Users\Admin>mongosh
Current Mongosh Log ID: 69b98e66111d7c69fb7c2906
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:          8.2.5
Using Mongosh:          2.7.0
mongosh 2.8.1 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

------
   The server generated these startup warnings when booting
   2026-03-16T17:56:05.598+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use cdac;
switched to db cdac
cdac> show databases;
admin   40.00 KiB
config  72.00 KiB
local   40.00 KiB
cdac> |
```

2. Display the list of collections in the CDAC database.

```
mongosh mongodb://127.0.0.
[d5_96525_Deven@sales]>CREATE INDEX cnum_odate_idx
    -> ON orders(cnum,odate);
Query OK, 0 rows affected (0.733 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>exit;
Bye

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   The server generated these startup warnings when booting
   2026-03-16T17:56:05.598+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
------

test> use cdac;
switched to db cdac
cdac> show databases;
admin   40.00 KiB
config  72.00 KiB
local   40.00 KiB
cdac> show collections;

cdac> |
```

3. Create a collection named LIBRARY.

```
cdac> db.createCollection('LIBRARY')
{ ok: 1 }
cdac> |
```

4. Insert a document into LIBRARY with: title: MongoDB programming, author: Amit, likes: 100

```
cdac> db.LIBRARY.find();
[
  {
    _id: ObjectId('69b9902c111d7c69fb7c2907'),
    title: 'MongoDB_programming',
    author: 'Amit',
    likes: 100
  }
]
cdac> |
```

5. Display the inserted document and note the _id field.

```
cdac> db.LIBRARY.find();
[
  {
    _id: ObjectId('69b9902c111d7c69fb7c2907'),
    title: 'MongoDB_programming',
    author: 'Amit',
    likes: 100
  }
]
cdac> |
```

6. Insert another document into LIBRARY with: title: MySQL programming, authors: [Jack, Jill], likes: 200

```
cdac> db.LIBRARY.insertOne({title: 'MySQL_programming', authors: ['Jack', 'Jill'], likes: 200 });
{
  acknowledged: true,
  insertedId: ObjectId('69b9985a111d7c69fb7c2909')
}
cdac> |
```

7. Display all documents in the LIBRARY collection.

```
cdac> db.LIBRARY.find();
[
  {
    _id: ObjectId('69b9902c111d7c69fb7c2907'),
    title: 'MongoDB_programming',
    author: 'Amit',
    likes: 100
  },
  {
    _id: ObjectId('69b9985a111d7c69fb7c2909'),
    title: 'MySQL_programming',
    authors: [ 'Jack', 'Jill' ],
    likes: 200
  }
]
cdac> |
```

8. Display only the first inserted document.

```
cdac> db.LIBRARY.find().limit(1);
[
  {
    _id: ObjectId('69b9902c111d7c69fb7c2907'),
    title: 'MongoDB_programming',
    author: 'Amit',
    likes: 100
  }
]
cdac> |
```

9. Display documents using the pretty() method.

```
cdac> db.LIBRARY.find().pretty();
[
  {
    _id: ObjectId('69b9902c111d7c69fb7c2907'),
    title: 'MongoDB_programming',
    author: 'Amit',
    likes: 100
  },
  {
    _id: ObjectId('69b9985a111d7c69fb7c2909'),
    title: 'MySQL_programming',
    authors: [ 'Jack', 'Jill' ],
    likes: 200
  }
]
cdac> |
```

10. Update the document where author = Amit to Amit Sharma.

```
cdac> db.LIBRARY.updateOne(
|   { author: 'Amit' },
|   { $set: { author: 'Amit Sharma' } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
cdac> |
```

11. Delete all documents with likes = 100.

```
cdac> db.LIBRARY.updateOne(
|   { author: 'Amit' },
|   { $set: { author: 'Amit Sharma' } }
| );
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
cdac> db.LIBRARY.deleteOne(
| {likes: 200}
| );
{ acknowledged: true, deletedCount: 1 }
cdac> |
```

12. Drop the LIBRARY collection.

```
cdac> db.LIBRARY.drop;
[Function: drop] AsyncFunction {
  apiVersions: [ 1, Infinity ],
  returnsPromise: true,
  serverVersions: [ '0.0.0', '999.999.999' ],
  topologies: [ 'ReplSet', 'Standalone', 'Sharded', 'LoadBalanced' ],
  returnType: { type: 'unknown', attributes: {} },
  deprecated: false,
  platforms: [ 'Compass', 'Browser', 'CLI' ],
  isDirectShellCommand: false,
  acceptsRawInput: false,
  shellCommandCompleter: undefined,
  newShellCommandCompleter: undefined,
  help: [Function (anonymous)] Help
}
cdac> |
```

13. Drop the CDAC database.

```
cdac> db.dropDatabase
[Function: dropDatabase] AsyncFunction {
  apiVersions: [ 1, Infinity ],
  returnsPromise: true,
  serverVersions: [ '0.0.0', '999.999.999' ],
  topologies: [ 'ReplSet', 'Standalone', 'Sharded', 'LoadBalanced' ],
  returnType: { type: 'unknown', attributes: {} },
  deprecated: false,
  platforms: [ 'Compass', 'Browser', 'CLI' ],
  isDirectShellCommand: false,
  acceptsRawInput: false,
  shellCommandCompleter: undefined,
  newShellCommandCompleter: undefined,
  help: [Function (anonymous)] Help
}
cdac>
```